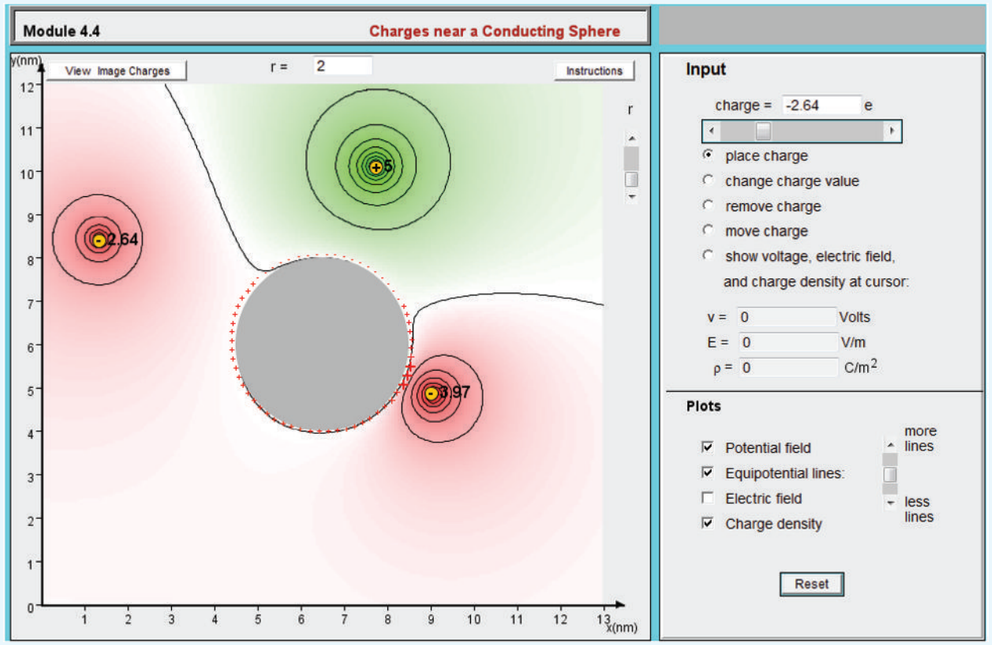
Module 4.4 Charges near a Conducting Sphere This module is similar to Module 4.3, except that now the conducting body is a sphere of selectable size.



If the material between the conductors is not a perfect dielectric (i.e., if it has a small conductivity σ), then current can flow through the material between the conductors, and the material exhibits a resistance R. The general expression for R for a resistor of arbitrary shape is given by Eq. (4.81):

$$R = \frac{-\int_l \mathbf{E}\cdot d\mathbf{l}}{\int_S \sigma\mathbf{E}\cdot d\mathbf{s}} \quad (\Omega). \tag{4.120}$$

For a medium with uniform σ and ε, the product of Eqs. (4.119) and (4.120) gives

$$RC = \frac{\varepsilon}{\sigma}. \tag{4.121}$$

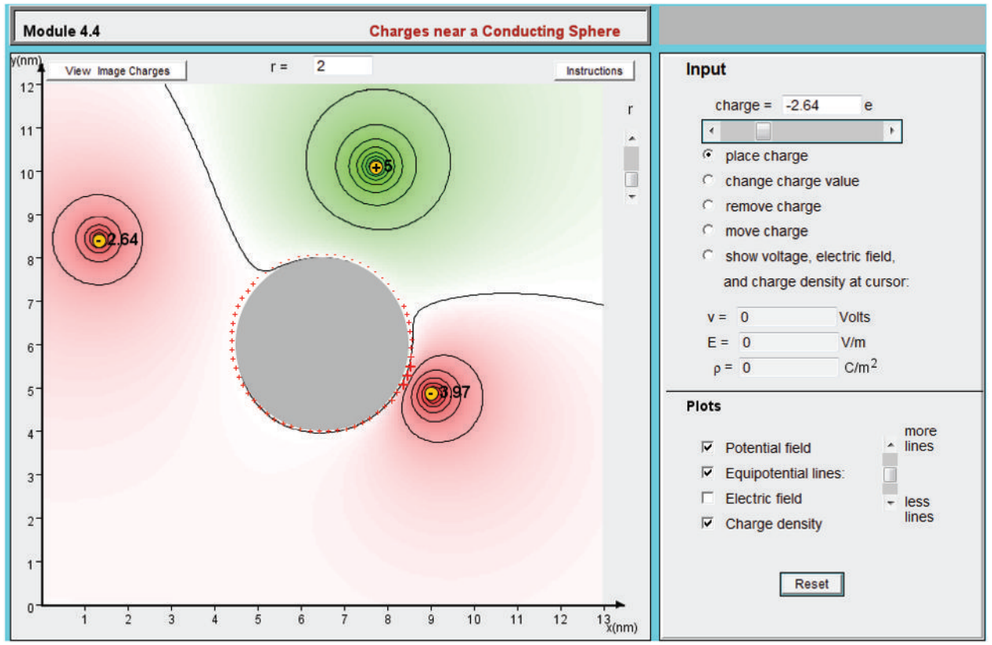
This simple relation allows us to find R if C is known, and vice versa.

Example 4-14: Capacitance of Parallel-Plate Capacitor

Obtain an expression for the capacitance C of a parallel-plate capacitor comprised of two parallel plates each of surface area A and separated by a distance d. The capacitor is filled with a dielectric material with permittivity ε.

Solution: In Fig. 4-26, we place the lower plate of the capacitor in the x–y plane and the upper plate in the plane $z = d$. Because of the applied voltage difference V, charges $+Q$ and $-Q$ accumulate on the top and bottom capacitor plates. If the plate dimensions are much larger than the separation d, then these charges distribute themselves quasi-uniformly across the plates, giving rise to a quasi-uniform field between them pointing in the $-\hat{\mathbf{z}}$ direction. In addition, a ***fringing field*** will exist near the capacitor edges, but its effects may be ignored because the bulk of the electric field exists between the plates.

模块 4.4 导电球体附近的电荷 该模块与模块 4.3 类似，不同之处在于现在导电体是可选择尺寸的球体。



如果导体之间的材料不是理想的电介质（即，如果它的电导率较小σ），则电流可以流过导体之间的材料，并且该材料表现出电阻R。任意形状的电阻器的R的一般表达式由式（1）给出。 （4.81）：

$$R = \frac{-\int_l \mathbf{E}\cdot d\mathbf{l}}{\int_S \sigma\mathbf{E}\cdot d\mathbf{s}} \quad (\Omega). \tag{4.120}$$

对于具有均匀 σ 和 ε 的介质，方程的乘积： (4.119) 和 (4.120) 给出

$$RC = \frac{\varepsilon}{\sigma}. \tag{4.121}$$

如果 C 已知，这个简单的关系允许我们找到 R，反之亦然。

例 4-14： 平行板电容器的电容

获得由两个平行板组成的平行板电容器的电容 C 的表达式，每个平行板的表面积为 A，且间隔距离 d。电容器充满介电常数为 ε 的介电材料。

解：在图4-26中，我们将电容器的下极板放置在x-y平面上，将上极板放置在平面z = d上。由于施加的电压差 V，电荷 +Q 和 −Q 积聚在顶部和底部电容器板上。如果板尺寸远大于间距 d，则这些电荷会准均匀地分布在板上，从而在它们之间产生指向 −ẑ 方向的准均匀场。此外，电容器边缘附近将存在边缘场，但其影响可以忽略不计，因为大部分电场存在于极板之间。

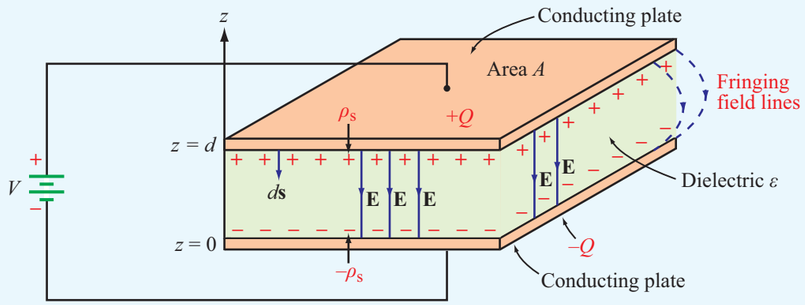


Figure 4-26 A dc voltage source connected to a parallel-plate capacitor (Example 4-14).

The charge density on the upper plate is $\rho_s = Q/A$. Hence, in the dielectric medium

$$\mathbf{E} = -\hat{\mathbf{z}}E,$$

and from Eq. (4.116), the magnitude of **E** at the conductor–dielectric boundary is $E = \rho_s/\varepsilon = Q/\varepsilon A$. From Eq. (4.118), the voltage difference is

$$V = -\int_0^d \mathbf{E}\cdot d\mathbf{l} = -\int_0^d (-\hat{\mathbf{z}}E)\cdot\hat{\mathbf{z}}\,dz = Ed, \qquad (4.122)$$

and the capacitance is

$$C = \frac{Q}{V} = \frac{Q}{Ed} = \frac{\varepsilon A}{d}, \qquad (4.123)$$

where use was made of the relation $E = Q/\varepsilon A$.

Example 4-15: Capacitance per Unit Length of Coaxial Line

Obtain an expression for the capacitance of the coaxial line shown in **Fig. 4-27**.

Solution: For a given voltage V across the capacitor, charges $+Q$ and $-Q$ accumulate on the surfaces of the outer and inner conductors, respectively. We assume that these charges are uniformly distributed along the length and circumference of the conductors with surface charge density $\rho_s' = Q/2\pi bl$ on the outer conductor and $\rho_s'' = -Q/2\pi al$ on the inner one. Ignoring fringing fields near the ends of the coaxial line, we can construct a cylindrical Gaussian surface in the dielectric in between the conductors with the radius r such that $a < r < b$. Symmetry implies that the **E**-field is identical at all points on this surface, which is directed radially inward. From Gauss's law, it follows that the field magnitude equals the absolute

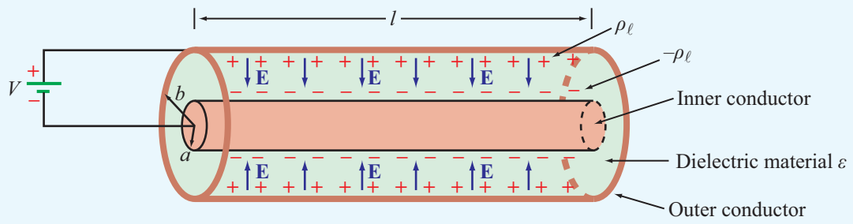


Figure 4-27 Coaxial capacitor filled with insulating material of permittivity ε (Example 4-15).

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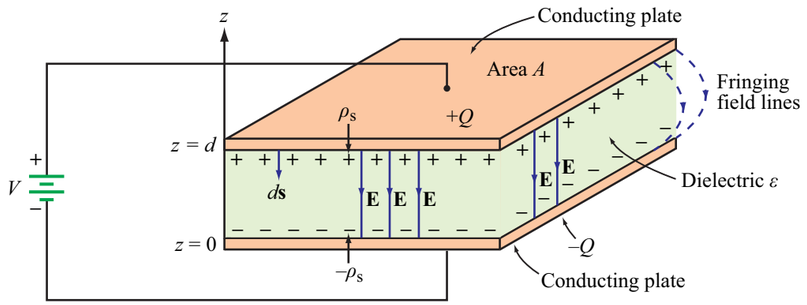


图 4-26 连接到平行板电容器的直流电压源（示例 4-14）。

上板上的电荷密度为$\rho_s = Q/A$。因此，在介电介质中

$$\mathbf{E} = -\hat{\mathbf{z}}E,$$

并从方程式根据方程（4.116），导体-电介质边界处的 E 大小为 $E = \rho_s/\varepsilon = Q/\varepsilon A$。从方程。（4.118），电压差为

$$V = -\int_0^d \mathbf{E}\cdot d\mathbf{l} = -\int_0^d (-\hat{\mathbf{z}}E)\cdot\hat{\mathbf{z}}\,dz = Ed, \qquad (4.122)$$

电容为

$$C = \frac{Q}{V} = \frac{Q}{Ed} = \frac{\varepsilon A}{d}, \qquad (4.123)$$

其中使用了关系 $E = Q/\varepsilon A$。

例 4-15： 同轴线单位长度的电容

求出图 4-27 所示同轴线电容的表达式。

解决方案：对于给定的电容器两端电压 V，电荷 +Q 和 −Q 分别积聚在外导体和内导体的表面上。我们假设这些电荷沿着导体的长度和圆周均匀分布，外导体上的表面电荷密度为 $\rho_s' = Q/2\pi bl$，内导体上的表面电荷密度为 $\rho_s'' = -Q/2\pi al$。忽略同轴线末端附近的边缘场，我们可以在导体之间的电介质中构造一个半径为 r 的圆柱形高斯表面，使得 $a < r < b$。对称性意味着该表面上所有点的电场都是相同的，并且径向向内指向。根据高斯定律，场强等于绝对场强

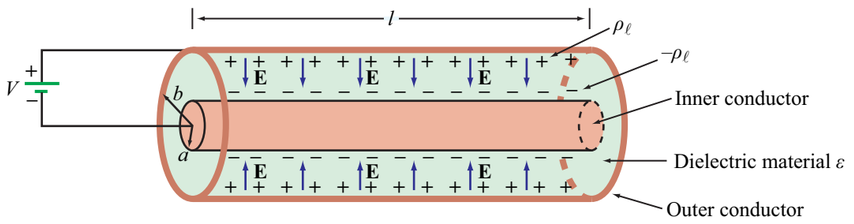


图4-27 填充介电常数为ε (的绝缘材料的同轴电容器示例4-15)。

value of the total charge enclosed, divided by the surface area. That is,

$$\mathbf{E} = -\hat{\mathbf{r}}\,\frac{Q}{2\pi\varepsilon r l}. \tag{4.124}$$

The potential difference V between the outer and inner conductors is

$$V = -\int_a^b \mathbf{E}\cdot d\mathbf{l} = -\int_a^b \left(-\hat{\mathbf{r}}\,\frac{Q}{2\pi\varepsilon r l}\right)\cdot(\hat{\mathbf{r}}\,dr)$$
$$= \frac{Q}{2\pi\varepsilon l}\ln\left(\frac{b}{a}\right). \tag{4.125}$$

The capacitance C is then given by

$$C = \frac{Q}{V} = \frac{2\pi\varepsilon l}{\ln(b/a)}, \tag{4.126}$$

and the capacitance per unit length of the coaxial line is

$$C' = \frac{C}{l} = \frac{2\pi\varepsilon}{\ln(b/a)} \quad \text{(F/m)}. \tag{4.127}$$

Concept Question 4-24: How is the capacitance of a two-conductor structure related to the resistance of the insulating material between the conductors?

Concept Question 4-25: What are fringing fields and when may they be ignored?

4-10 Electrostatic Potential Energy

A source connected to a capacitor expends energy in charging up the capacitor. If the capacitor plates are made of a good conductor with effectively zero resistance, and if the dielectric separating the two plates has negligible conductivity, then no real current can flow through the dielectric, and no ohmic losses occur anywhere in the capacitor. Where then does the energy expended in charging up the capacitor go? The energy ends up getting stored in the dielectric medium in the form of ***electrostatic potential energy***. The amount of stored energy W_e is related to Q, C, and V.

Suppose we were to charge up a capacitor by ramping up the voltage across it from $\upsilon = 0$ to $\upsilon = V$. During the process, charge $+q$ accumulates on one conductor and $-q$ on the other. In effect, a charge q has been transferred from one of the conductors to the other. The voltage υ across the capacitor is related to q by

$$\upsilon = \frac{q}{C}. \tag{4.128}$$

From the definition of υ, the amount of work dW_e required to transfer an additional incremental charge dq from one conductor to the other is

$$dW_e = \upsilon\, dq = \frac{q}{C}\,dq. \tag{4.129}$$

If we transfer a total charge Q between the conductors of an initially uncharged capacitor, then the total amount of work performed is

$$W_e = \int_0^Q \frac{q}{C}\,dq = \frac{1}{2}\,\frac{Q^2}{C} \quad \text{(J)}. \tag{4.130}$$

Using $C = Q/V$, where V is the final voltage, W_e also can be expressed as

$$W_e = \tfrac{1}{2}CV^2 \quad \text{(J)}. \tag{4.131}$$

The capacitance of the parallel-plate capacitor discussed in Example 4-14 is given by Eq. (4.123) as $C = \varepsilon A/d$, where A is the surface area of each of its plates and d is the separation between them. Also, the voltage V across the capacitor is related to the magnitude of the electric field E in the dielectric by $V = Ed$. Using these two expressions in Eq. (4.131) gives

$$W_e = \tfrac{1}{2}\,\frac{\varepsilon A}{d}(Ed)^2 = \tfrac{1}{2}\,\varepsilon E^2(Ad) = \tfrac{1}{2}\,\varepsilon E^2\upsilon, \tag{4.132}$$

where $\upsilon = Ad$ is the volume of the capacitor. This expression affirms the assertion made at the beginning of this section, namely that the energy expended in charging up the capacitor is being stored in the electric field present in the dielectric material in between the two conductors.

The ***electrostatic energy density*** w_e is defined as the electrostatic potential energy W_e per unit volume:

$$w_e = \frac{W_e}{\upsilon} = \frac{1}{2}\,\varepsilon E^2 \quad \text{(J/m}^3\text{)}. \tag{4.133}$$

Even though this expression was derived for a parallel-plate capacitor, it is equally valid for any dielectric medium containing an electric field $\mathbf{E}$, including a vacuum. Furthermore, for any volume υ, the total electrostatic potential energy stored in it is

$$W_e = \frac{1}{2}\int_\upsilon \varepsilon E^2\, d\upsilon \quad \text{(J)}. \tag{4.134}$$

Returning to the parallel-plate capacitor, the oppositely charged plates are attracted to each other by an ***electrical force*** $\mathbf{F}_e$. In terms of the coordinate system of Fig. 4-28, the

所包含的总电荷除以表面积的值。那是，

$$\mathbf{E} = -\hat{\mathbf{r}}\,\frac{Q}{2\pi\varepsilon r l}. \tag{4.124}$$

外导体和内导体之间的电位差 V 为

$$V = -\int_a^b \mathbf{E}\cdot d\mathbf{l} = -\int_a^b \left(-\hat{\mathbf{r}}\,\frac{Q}{2\pi\varepsilon r l}\right)\cdot(\hat{\mathbf{r}}\,dr)$$
$$= \frac{Q}{2\pi\varepsilon l}\ln\left(\frac{b}{a}\right). \tag{4.125}$$

电容 C 由下式给出

$$C = \frac{Q}{V} = \frac{2\pi\varepsilon l}{\ln(b/a)}, \tag{4.126}$$

同轴线单位长度的电容为

$$C' = \frac{C}{l} = \frac{2\pi\varepsilon}{\ln(b/a)} \quad \text{(F/m)}. \tag{4.127}$$

概念问题 4-24：双导体结构的电容与导体之间绝缘材料的电阻有何关系？

概念问题 4-25：什么是边缘场以及何时可以忽略它们？

4-10 静电势能

连接到电容器的电源在给电容器充电时消耗能量。如果电容器极板由有效为零电阻的良导体制成，并且分隔两个极板的电介质的电导率可以忽略不计，则没有实际电流可以流过电介质，并且电容器中的任何地方都不会发生欧姆损耗。那么电容器充电所消耗的能量去了哪里呢？能量最终以静电势能的形式存储在电介质中。储存的能量 W_e 与 Q、C 和 V 相关。

假设我们要通过将电容器两端的电压从 $\upsilon = 0$ 提升到 $\upsilon = V$ 来给电容器充电。在此过程中，电荷 +q 积聚在一个导体上，而 −q 积聚在另一导体上。实际上，电荷 q 已从一个导体转移到另一个导体。电容器两端的电压 υ 与 q 的关系为

$$\upsilon = \frac{q}{C}. \tag{4.128}$$

根据 υ 的定义，将额外增量电荷 dq 从一个导体转移到另一个导体所需的功 dW_e 为

$$dW_e = \upsilon\, dq = \frac{q}{C}\,dq. \tag{4.129}$$

如果我们在最初未充电的电容器的导体之间转移总电荷 Q，则所做的总功为

$$W_e = \int_0^Q \frac{q}{C}\,dq = \frac{1}{2}\,\frac{Q^2}{C} \quad \text{(J)}. \tag{4.130}$$

使用 C = Q/V，其中 V 是最终电压，W_e 也可以表示为

$$W_e = \tfrac{1}{2}CV^2 \quad \text{(J)}. \tag{4.131}$$

例 4-14 中讨论的平行板电容器的电容由公式 1 给出。(4.123) 为 C = ε A/d，其中 A 是每个板的表面积，d 是它们之间的间距。此外，电容器两端的电压 V 与电介质中电场 E 的大小有关，关系为 V = Ed。在等式中使用这两个表达式。(4.131) 给出

$$W_e = \tfrac{1}{2}\,\frac{\varepsilon A}{d}(Ed)^2 = \tfrac{1}{2}\,\varepsilon E^2(Ad) = \tfrac{1}{2}\,\varepsilon E^2\upsilon, \tag{4.132}$$

其中 υ = Ad 是电容器的体积。这个表达式证实了本节开头的断言，即电容器充电所消耗的能量被存储在两个导体之间的介电材料中存在的电场中。

静电能量密度 w_e 定义为每单位体积的静电势能 W_e：

$$w_e = \frac{W_e}{\upsilon} = \frac{1}{2}\,\varepsilon E^2 \quad \text{(J/m}^3\text{)}. \tag{4.133}$$

尽管这个表达式是针对平行板电容器导出的，但它对于任何包含电场 E 的介电介质（包括真空）同样有效。此外，对于任何体积υ，其中存储的总静电势能是

$$W_e = \frac{1}{2}\int_\upsilon \varepsilon E^2\, d\upsilon \quad \text{(J)}. \tag{4.134}$$

回到平行板电容器，带相反电荷的板通过电力 F_e 相互吸引。就图4-28的坐标系而言，

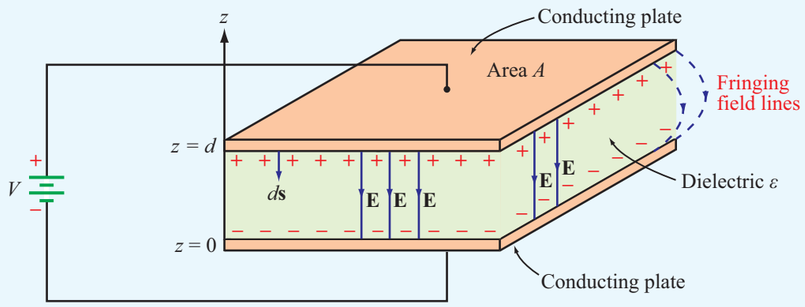


Figure 4-28 A dc voltage source connected to a parallel-plate capacitor.

electrical force acting on the upper plate is along $-\hat{\mathbf{z}}$ (due to attraction by the lower plate). Hence, it is given by

$$\mathbf{F}_e = -\hat{\mathbf{z}}\, F_e \qquad \text{(force on upper plate).} \tag{4.135}$$

Our plan is to compute $\mathbf{F}_e$ from energy considerations. We start by converting the spacing d into a variable z and using $C = \varepsilon A/z$ in Eq. (4.131):

$$W_e = \tfrac{1}{2} CV^2 = \tfrac{1}{2} \frac{\varepsilon A V^2}{z}. \tag{4.136}$$

If V is maintained at a constant level, W_e decreases when increasing the separation z between the plates. If an external, upward-directed force $\mathbf{F} = -\mathbf{F}_e$ is applied to counter the electrostatic force $\mathbf{F}_e$ and used to move the upper plate upwards by a distance dz, the expended mechanical work is

$$dW = \mathbf{F} \cdot \hat{\mathbf{z}}\, dz. \tag{4.137}$$

The work dW is equal to the loss in electrostatic energy stored in the capacitor. That is,

$$dW = -dW_e. \tag{4.138}$$

Also, $\mathbf{F}_e = -\mathbf{F}$, which leads to

$$dW_e = \mathbf{F}_e \cdot \hat{\mathbf{z}}\, dz = -\hat{\mathbf{z}}\, F_e \cdot \hat{\mathbf{z}}\, dz = -F_e\, dz. \tag{4.139}$$

From Eq. (4.136),

$$dW_e = -\tfrac{1}{2}\, \varepsilon \frac{AV^2}{z^2}\, dz. \tag{4.140}$$

Equating Eqs. (4.139) and (4.140) and replacing z with d leads to

$$F_e = \tfrac{1}{2}\, \varepsilon \frac{AV^2}{d^2}, \tag{4.141a}$$

and

$$\mathbf{F}_e = -\hat{\mathbf{z}}\, \tfrac{1}{2}\, \varepsilon A \frac{V^2}{d^2} \qquad \text{(N).} \tag{4.141b}$$

(parallel-plate capacitor)

This is the electrostatic force exerted on the upper plate. The force on the lower plate is identical in magnitude and opposite in direction.

The relationship given by Eq. (4.139) pertains to a capacitor with $d\mathbf{l} = \hat{\mathbf{z}}\, dz$. We can generalize the result for $d\mathbf{l}$ along any direction as

$$\mathbf{F}_e = -\nabla W_e. \tag{4.142}$$

Example 4-16: Force on Sliding Dielectric

The two plates of the parallel-plate capacitor shown in Fig. 4-29 are each of length ℓ and width w, and the separation between them is d. The capacitor contains a dielectric block of dimensions $\ell \times w \times d$ and permittivity ε. The block can slide in and out of the capacitor cavity along its length dimension. Compute the force $\mathbf{F}_e$ acting on the dielectric block when it is partially outside of the cavity and the voltage across the capacitor is V.

Solution: From Eq. (4.122), the electric field inside the capacitor cavity is

$$E = \frac{V}{d}.$$

This is true in both the section containing the dielectric block and the section filled with air. The total electrostatic energy of

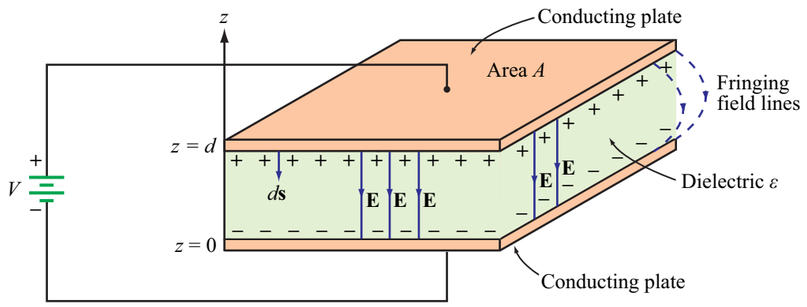


图 4-28 连接到平行板电容器的直流电压源。

作用在上板上的电力沿−ẑ（由于下板的吸引力）。因此 和 ，它由下式给出

$$\mathbf{F}_e = -\hat{\mathbf{z}}\, F_e \qquad \text{(force on upper plate).} \tag{4.135}$$

我们的计划是从能量考虑来计算 F_e。我们首先将间距 d 转换为变量 z 并在等式中使用 $C = \varepsilon A/z$。 （4.131）：

$$W_e = \tfrac{1}{2} CV^2 = \tfrac{1}{2} \frac{\varepsilon A V^2}{z}. \tag{4.136}$$

如果 V 保持在恒定水平，则当增加板之间的间距 z 时，W_e 会减小。如果施加一个向上的外部力 $F = -F_e$ 来抵消静电力 F_e 并将上板向上移动距离 dz，则消耗的机械功为

$$dW = \mathbf{F} \cdot \hat{\mathbf{z}}\, dz. \tag{4.137}$$

功 dW 等于电容器中存储的静电能的损失。那是，

$$dW = -dW_e. \tag{4.138}$$

此外，$F_e = -F$，这导致

$$dW_e = \mathbf{F}_e \cdot \hat{\mathbf{z}}\, dz = -\hat{\mathbf{z}}\, F_e \cdot \hat{\mathbf{z}}\, dz = -F_e\, dz. \tag{4.139}$$

从方程。 (4.136),

$$dW_e = -\tfrac{1}{2}\, \varepsilon \frac{AV^2}{z^2}\, dz. \tag{4.140}$$

等式(4.139) 和 (4.140) 并将 z 替换为 d 导致

$$F_e = \tfrac{1}{2}\, \varepsilon \frac{AV^2}{d^2}, \tag{4.141a}$$

$$\mathbf{F}_e = -\hat{\mathbf{z}}\, \tfrac{1}{2}\, \varepsilon A \frac{V^2}{d^2} \qquad \text{(N).} \tag{4.141b}$$

(平行板电容器)

这是施加在上板上的静电力。下板上的力大小相同，方向相反。

由方程式给出的关系(4.139) 涉及具有 $d\mathbf{l} = \hat{\mathbf{z}}\, dz$ 的电容器。我们可以将 dl 沿任意方向的结果概括为

$$\mathbf{F}_e = -\nabla W_e. \tag{4.142}$$

例 4-16： 滑动电介质上的力

图4-29所示的平行板电容器的两个极板的长度为ℓ，宽度为w，它们之间的间距为d。电容器包含尺寸为 $\ell \times w \times d$ 和介电常数为 ε 的介电块。该块可以沿着其长度尺寸滑入和滑出电容器腔。计算当介电块部分位于腔体外部并且电容器两端的电压为 V 时作用在介电块上的力 F_e。

解决方案：从方程。 （4.122），电容器腔内的电场为

$$E = \frac{V}{d}.$$

这对于包含介电块的部分和充满空气的部分都是如此。总静电能量为

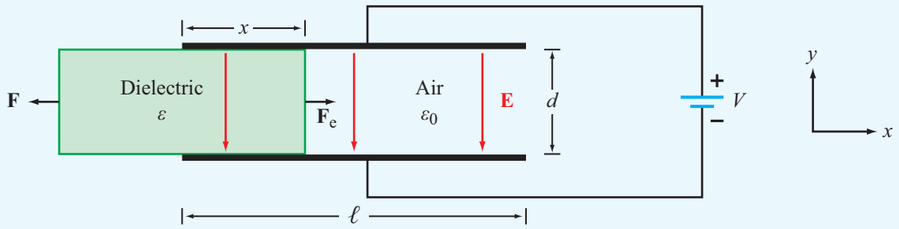


Figure 4-29 Parallel-plate capacitor with slidable dielectric block.

the capacitor consists of two components: one for the volume containing the dielectric block of permittivity ε and volume $\upsilon_1 = xwd$, and another for the volume containing air with ε_0 and volume $\upsilon_2 = (\ell - x)wd$. Hence,

$$\begin{aligned} W_e &= \tfrac{1}{2}\varepsilon E^2 \upsilon_1 + \tfrac{1}{2}\varepsilon_0 E^2 \upsilon_2 \\ &= \tfrac{1}{2}\varepsilon \left(\frac{V}{d}\right)^2 xwd + \tfrac{1}{2}\varepsilon_0 \left(\frac{V}{d}\right)^2 (\ell - x)wd \\ &= \frac{1}{2}\frac{V^2}{d} w[\varepsilon x + \varepsilon_0(\ell - x)]. \end{aligned} \tag{4.143}$$

Since $\varepsilon > \varepsilon_0$, the electrostatic energy is maximum when $x = \ell$ (dielectric block fully inside the cavity). Sliding the dielectric block out of the capacitor requires exerting an external mechanical force $\mathbf{F}$ to oppose the electrostatic force $\mathbf{F}_e$, whose tendency is to oppose reduction in W_e. Thus, the direction of $\mathbf{F}_e$ is to pull the block back into the capacitor.

The magnitude of $\mathbf{F}_e$ can be obtained from

$$\begin{aligned} F_e &= \frac{dW_e}{dx} \\ &= \frac{d}{dx}\left[\frac{1}{2}\frac{V^2}{d} w[\varepsilon x + \varepsilon_0(\ell - x)]\right] \\ &= \frac{1}{2}\frac{V^2}{d} w(\varepsilon - \varepsilon_0). \end{aligned} \tag{4.144}$$

Concept Question 4-26: To bring a charge q from infinity to a given point in space, a certain amount of work W is expended. Where does the energy corresponding to W go?

Concept Question 4-27: When a voltage source is connected across a capacitor, what is the direction of the electrical force acting on its two conducting surfaces?

Exercise 4-18: The radii of the inner and outer conductors of a coaxial cable are 2 cm and 5 cm, respectively, and the insulating material between them has a relative permittivity of 4. The charge density on the outer conductor is $\rho_\ell = 10^{-4}$ (C/m). Use the expression for $\mathbf{E}$ derived in Example 4-15 to calculate the total energy stored in a 20 cm length of the cable.

Answer: $W_e = 4.1$ J. (See EM.)

4-11 Image Method

Consider a point charge Q at a distance d above a horizontally infinite, perfectly conducting plate (Fig. 4-30(a)). We want to determine V and $\mathbf{E}$ at any point in the space above the plate, as well as the surface charge distribution on the plate. Three different methods for finding $\mathbf{E}$ have been introduced in this chapter The first method, based on Coulomb's law, requires knowledge of the magnitudes and locations of all the charges. In the present case, the charge Q induces an unknown and nonuniform distribution of charge on the plate. Hence, we cannot utilize Coulomb's method. The second method, based on Gauss's law, is equally difficult to use because it is not clear how to construct a Gaussian surface across which $\mathbf{E}$ is only tangential or only normal. The third method is based on evaluating the electric field using $\mathbf{E} = -\nabla V$ after solving Poisson's or Laplace's equation for V subject to the available boundary conditions, but it is mathematically involved.

Alternatively, the problem at hand can be solved using *image theory*.

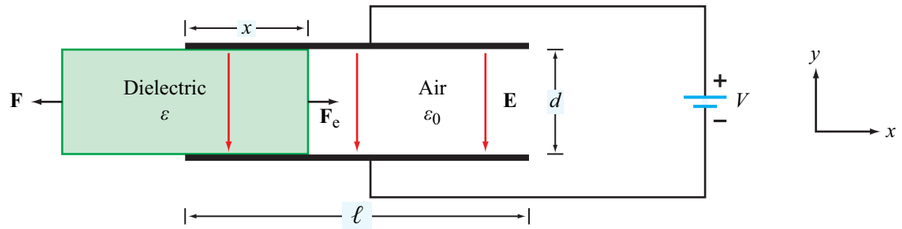


图 4-29 带有可滑动电介质块的平行板电容器。

该电容器由两部分组成：一个部分为包含介电常数为 ε 且体积为 $\upsilon_1 = xwd$ 的介电块的体积，另一个部分为包含具有 ε_0 和体积 $\upsilon_2 = (\ell - x)wd$ 的空气的体积。因此，

$$\begin{aligned} W_e &= \tfrac{1}{2}\varepsilon E^2 \upsilon_1 + \tfrac{1}{2}\varepsilon_0 E^2 \upsilon_2 \\ &= \tfrac{1}{2}\varepsilon \left(\frac{V}{d}\right)^2 xwd + \tfrac{1}{2}\varepsilon_0 \left(\frac{V}{d}\right)^2 (\ell - x)wd \\ &= \frac{1}{2}\frac{V^2}{d} w[\varepsilon x + \varepsilon_0(\ell - x)]. \end{aligned} \tag{4.143}$$

自从$\varepsilon > \varepsilon_0$，当 x 时静电能最大$= \ell$ (介电块完全位于空腔内)。将介电块滑出电容器需要施加外部机械力 F 来抵抗静电力 F_e，其倾向是反对减少 W_e。因此，F 的方向e是将块拉回到电容器中。

F_e 的大小可由下式获得

$$\begin{aligned} F_e &= \frac{dW_e}{dx} \\ &= \frac{d}{dx}\left[\frac{1}{2}\frac{V^2}{d} w[\varepsilon x + \varepsilon_0(\ell - x)]\right] \\ &= \frac{1}{2}\frac{V^2}{d} w(\varepsilon - \varepsilon_0). \end{aligned} \tag{4.144}$$

概念问题 4-26：为了将电荷 q 从无穷大带到空间中的给定点，需要消耗一定量的功 W。 W 对应的能量去了哪里？

概念问题 4-27：当电压源连接在电容器两端时，作用在其两个导电表面上的电力的方向是什么？

练习4-18： 同轴电缆的内导体和外导体的半径分别为 2 cm和5 cm，它们之间的绝缘材料的相对介电常数为 4。外导体上的电荷密度为$\rho_\ell = 10^{-4}$ (C/m)。使用例 4 -15 中导出的 E 表达式来计算 20 cm 长度的电缆中存储的总能量。

答案：$W_e = 4.1$ J.（见EM。）

4-11 镜像法

考虑在水平无限、理想导电的板上方距离 d 处的点电荷 Q（图 4-30(a)）。我们想要确定板上方空间中任意点的 V 和 E，以及板上的表面电荷分布。本章介绍了寻找 E 的三种不同方法。第一种方法基于库仑定律，需要了解所有电荷的大小和位置。在本例中，电荷 Q 在板上产生未知且不均匀的电荷分布。因此，我们不能使用库仑法。第二种方法基于高斯定律，同样难以使用，因为尚不清楚如何构造 E 仅为切线或法线的高斯曲面。第三种方法是在根据可用边界条件求解 V 的泊松或拉普拉斯方程后，使用 $\mathbf{E} = -\nabla V$ 评估电场，但它涉及数学。

或者，可以使用镜像理论来解决当前的问题。